

Registered Position-Force Sample Catalog

Stage 11E0b: compact species evidence, represented time, force provenance, and structural masks

mdstats

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Purpose and stage boundary

Revision-42 migration status

The current runtime baseline predates `SimulationControlCertificate`, `ProductionRegimeCatalog`, `PmfAdmissibilityCertificate`, and `EvidenceAdmissibilityOverlay`. Existing serialized catalogs may therefore still contain legacy scientific masks. Revision 42 defines the target migration: E0b retains raw availability and geometry masks only, while ENS/STAT construct signed overlays. Until that implementation stage is complete, legacy masks are compatibility data and cannot override unresolved ENS/STAT evidence.

Stage 11E0b creates the compact trajectory-evidence object consumed by later nonparametric site discovery. It joins the immutable source collection to one Stage-C0 registration without introducing a density estimator, site label, basin, transition state, or kinetic rate.

The canonical module is:

```
mdstats.analysis.site_samples
```

The scientific path is:

```
source collection
+ certified FrameRegistrationResult
+ declared temporal segments
```

- + optional topology regimes
- + force and thermodynamic provenance
 - > compact one-species sample catalog
 - > exact evidence-channel views
 - > later density / force / validation stages

The catalog never eagerly materializes every ion-frame-ring or ion-frame-tile descriptor. Structural annotations remain lazy and are resolved only for an explicitly requested sample subset after a candidate has been associated with a persistent structural identity.

Coordinate and identity contract

One catalog contains exactly one atomic number. Let the selected atoms be

$$A_Z = \{a_0, \dots, a_{N_Z-1}\}.$$

The compact sample index is frame-major:

$$s = tN_Z + j, \quad (t, j) \in \{0, \dots, N_t - 1\} \times \{0, \dots, N_Z - 1\}.$$

Each sample retains:

- source frame index and persistent frame ID;
- source atom index and species atomic number;
- registered unwrapped Cartesian position;
- registered wrapped fractional coordinate and integer image shift;
- transformed force covector when available;
- represented-time weight;
- exact topology-regime ID or the explicit unknown value -1; and
- all evidence masks and provenance signatures.

The authoritative registration products are copied only for the selected species. The catalog verifies independently that

$$q_{ta} = x_{ta}M_t + b_t$$

for the supplied source collection and registration. A registration produced from a different collection fails closed even when array shapes happen to match.

Segment-aware represented time

Continuous trajectory segments

A trajectory is divided into explicit continuous segments. Segment starts include:

- frame zero;
- user-declared restart or discontinuity boundaries; and
- Stage-C0 translation-branch reset points.

No represented-time interval crosses a segment boundary.

For times

$$t_0 < t_1 < \dots < t_{n-1}$$

inside one segment, the frame weights are the midpoint control-volume widths, equivalently the sample weights of composite trapezoidal quadrature:

$$w_0 = \frac{t_1 - t_0}{2},$$

$$w_i = \frac{t_{i+1} - t_{i-1}}{2}, \quad 0 < i < n - 1,$$

and

$$w_{n-1} = \frac{t_{n-1} - t_{n-2}}{2}.$$

A one-frame trajectory segment has no represented duration and is rejected unless explicit nonnegative frame weights are supplied.

Independent ensembles

Independent ensembles receive unit frame weights and every frame is a segment start. This is a sample-count measure, not a temporal-continuity claim.

Segment inclusion

A multi-segment trajectory requires explicit `included_segment_ids`. The library does not silently pool restarts, heating, equilibration, or production segments. Pooling a declared heating segment with a declared production segment requires a separate explicit opt-in.

Every segment has one of the labels:

heating
equilibration
production
other
unknown

The labels are provenance. They do not infer equilibrium or stationarity.

Topology regimes and structural evidence

`TopologyRegimeAssignment` retains per-frame:

- exact topology-regime ID;
- connectivity-flicker mask;
- structural-evidence mask; and
- optional source `TopologyCatalog` digest.

When an exact topology catalog is supplied:

- catalog frame IDs must match the source collection;
- exact topology IDs are projected to source-frame positions;
- transient topology segments are marked as connectivity flicker; and
- the two frames adjacent to each exact transition are marked as flicker by default.

A connectivity-flicker frame cannot simultaneously be accepted as structural evidence. Unknown topology IDs may still retain position evidence when the user has explicitly allowed the structural frame; the unknown identity remains visible rather than being replaced by a guessed class.

Nested evidence masks

For every compact sample, retain the source masks

$$P_0, \quad F_0,$$

the temporal and structural masks

$$T, \quad S,$$

and the connectivity-flicker diagnostic

$$C.$$

The effective position and force masks are defined exactly by

$$P = P_0 \wedge T \wedge S,$$

$$F = F_0 \wedge T \wedge S.$$

The matched density-force subset is

$$J = P \wedge F.$$

`joint_mask` must equal this intersection bit-for-bit. A matched force-density comparison obtains positions, forces, identities, topology regimes, and weights from the same compact sample indices. It is not permitted to recompute two independent selections and assume they match.

Force and PMF provenance

The catalog retains, without reinterpretation:

- source force completeness;
- bias or constraint-force evidence;
- stochastic or thermostat-force evidence;
- geometric force-transform status;
- PMF force-admissibility status;
- Stage-C0 registration signature; and
- explanatory reasons from the force-admissibility contract.

Ordinary geometric force evidence and thermodynamic PMF evidence are distinct. A diagnostic structure-fitted force projection may remain available in `F` and `J`, while remaining forbidden from the PMF subset.

The immutable `E0b` catalog stores raw availability and geometry masks. It does not construct a PMF subset. A later `EvidenceAdmissibilityOverlay` may define

$$J_{\text{PMF}} = J \wedge A_F \wedge R \wedge P,$$

where:

- J is the raw joint position/force availability mask;
- A_F is the Stage-C0 force-coordinate admissibility result;
- R selects a tested production regime from `ProductionRegimeCatalog`; and
- P is the relevant accepted permission in `PmfAdmissibilityCertificate`.

No user declaration or untested stationarity assertion can set R or P . Framewise temperatures remain provenance and do not become one PMF temperature implicitly. Unresolved prerequisite certificates produce an unresolved or empty overlay while preserving the raw catalog.

Sampling-state and temperature provenance

Source declarations, tested diagnostics, and promotion decisions remain separate:

```
source_declaration:
  user_comment_only
  explicit_protocol_claim
  unknown
```

```
diagnostic_status:
  resolved
  unresolved
  insufficient
  rejected
  unavailable
  not_applicable
```

```
promotion_decision:
  accepted
  conditional
```

blocked
rejected

Every overlay records the exact ensemble, production-regime, PMF-admissibility, policy, and E0b signatures.

PMFTemperatureProvenance distinguishes:

declared_constant
framewise_observed
unavailable
unknown

A constant PMF temperature must be positive and include a source declaration. An observed framewise range is never promoted automatically to a thermodynamic PMF declaration.

Evidence views and measures

The catalog exposes exact views for:

position
force
joint
pmf_force

Each view contains the selected compact indices, source frame/atom identities, registered positions, forces when required, represented-time weights, and regime IDs.

For channel K , the ion-time measure is

$$\mathcal{T}_K = \sum_{s \in K} w_s.$$

The normalized sample weights are

$$\hat{w}_s = \frac{w_s}{\mathcal{T}_K}.$$

Later density stages must state whether they use ion-time measure, normalized probability measure, or mean occupancy. Stage 11E0b stores the source measure but does not choose a density normalization.

Lazy structural annotations

LazyStructuralAnnotationView accepts a resolver callable and materializes annotations only for:

- one declared evidence channel; or
- one explicit compact sample-index sequence.

The resolver returns a mapping of arrays with leading dimension equal to the requested sample count. Results are immutable and cached by exact sample-index tuple. The catalog itself stores no eager $N_{\text{ion}} N_{\text{frame}} N_{\text{ring}}$ tensor.

The resolver is intentionally not serialized. It is executable analysis logic, not source evidence.

Multi-trajectory registration groups

FrameRegistrationGroup certifies that multiple trajectory registrations share one fixed periodic domain. Every member must have:

- identical periodic axes;
- one fixed registered cell within the declared tolerance;
- the same analysis-geometry metric digest; and
- a registered cell equal to the shared cell within tolerance.

The group retains member source-contract and registration signatures, internal cell deviations, cross-member deviation, and a group signature. A grouped catalog records its exact member index and group signature.

The group does not assert kinetic pooling, equilibrium equivalence, identical force provenance, or statistical exchangeability. Those are later validation questions.

Serialization and immutability

All persistent records use strict JSON schemas and deterministic SHA-256 signatures. Arrays become read-only copies at construction. Catalog signatures bind:

- compact identities and coordinates;
- transformed force values;
- time weights and regime IDs;
- every evidence-mask signature;
- temporal, topology, force, state, and temperature provenance;
- source, registration, policy, and optional registration-group signatures; and
- JSON-safe metadata.

A changed coordinate, mask, weight, provenance field, or signature fails replay. Lazy annotation resolvers and caches are not serialized.

Failure behavior

The stage fails closed for:

- no matching species atoms;
- mixed atomic numbers in one species catalog;
- a registration not bound geometrically to the collection;
- frame IDs that disagree across temporal or topology products;
- hidden pooling of multiple trajectory segments;
- heating/production pooling without explicit opt-in;
- non-increasing times inside a continuous segment;
- one-frame trajectory segments without explicit weights;
- selected force evidence without transformed forces;
- a connectivity-flicker sample accepted as structural evidence;
- a joint mask unequal to the exact position/force intersection;
- a PMF mask outside the joint subset;
- incompatible multi-trajectory registered domains; or
- malformed or tampered serialized records.

Public API

The principal entry points are:

```
prepare_trajectory_segment_weighting(...)
prepare_topology_regime_assignment(...)
prepare_frame_registration_group(...)
prepare_framework_aligned_ion_sample_catalog(...)
```

The principal records are:

```
TrajectorySegmentWeighting
TopologyRegimeAssignment
SamplingStateProvenance
PMFTemperatureProvenance
SampleForceProvenance
SampleEvidenceMasks
FrameRegistrationGroup
FrameworkAlignedIonSampleCatalog
SpeciesSampleEvidenceView
LazyStructuralAnnotationView
```

They are exported from `mdstats.analysis` and the package root.

Acceptance requirements

- Represented-time weights do not cross declared discontinuities.
- A multi-segment trajectory is not pooled without an explicit segment subset.
- Heating and production are not mixed silently.
- One catalog contains one species and the exact registered C0 coordinates.
- Position, force, joint, and PMF evidence use their declared immutable masks.
- `joint_mask` is exactly `position_mask & force_mask`.
- Matched position-force views use identical compact sample indices and weights.
- Connectivity-flicker frames remain explicit and are excluded from structural evidence.
- Diagnostic geometric forces can remain visible without becoming PMF evidence.
- PMF evidence requires force, equilibrium, stationarity, and constant-temperature provenance simultaneously.
- Structural annotations are lazy and subset-bound.
- Compatible registration groups certify one fixed periodic domain and reject mismatched cells, metrics, or periodic axes.
- Serialization replay is deterministic and tamper-evident.

Deferred work

Stage 11E0b does not implement:

- periodic kernel density estimation;
- density scores, metric gradients, or Hessians;
- attractors, ridges, basins, or scale selection;
- structural candidate association;
- M–O or M–T fingerprints;
- force-field reconstruction;
- residence segmentation;
- transition-path extraction; or
- rate estimation.

These remain Stage 11E1 and later boundaries.

Method provenance

The midpoint control-volume weights are standard composite-trapezoidal sampled data quadrature; SciPy documents the equivalent trapezoidal integration rule [1]. Stage 11E0b’s segment resets, explicit inclusion gate, compact evidence catalog, exact nested-mask algebra, conservative PMF subset, registration-group certificate, and lazy structural annotation boundary are project-specific constructions.

Force covector transformation and PMF admissibility are not re-derived here; they are inherited from the Stage-C0A1/C0A2 coordinate specifications.

Reference

[1] SciPy Developers. `scipy.integrate.trapezoid` documentation. <https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.trapezoid.html>.